

- Correction: Not completely relevant but in the lecture I wrote

$$E_n = \frac{\hbar^2 \pi^2}{2^2} (n_x^2 + n_y^2 + n_z^2)$$

this should be

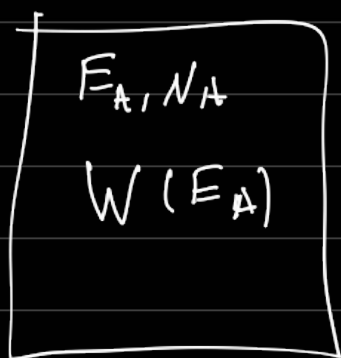
$$E_n = \frac{\hbar^2 \pi^2}{2m L^2} (n_x^2 + n_y^2 + n_z^2)$$

- What I wrote is really p^2 since

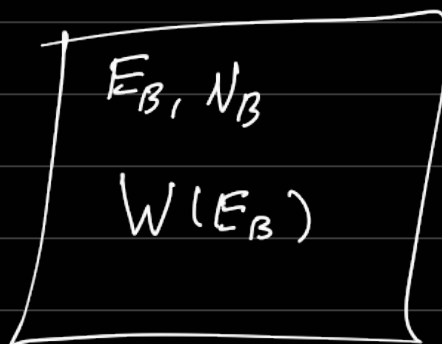
$$E = p^2/2m \Rightarrow p^2 = 2mE.$$

* Not completely relevant to discussion but worth mentioning.

Recap: Two systems



A



B

$$W(E_A, E_B) = W(E_A) W(E_B)$$

↳ System before interacting

Found: [5.2.3] $\frac{1}{T} = \left(\frac{\partial S}{\partial E} \right)$

- At equilibrium all states are equally probable
- We then let the systems interact, at which point one system 'gives' energy

$$\sim dE_B = dE_A$$

~ Maximized $W(E_A, E_B)$:

$$\frac{\partial W}{\partial E_A}$$

$$\frac{\partial E}{\partial E} \bigg|_{E=E_*} = 0$$

~> This means $T_B > T_A$. [5.2.6]

* Okay, so lets add to all of this.

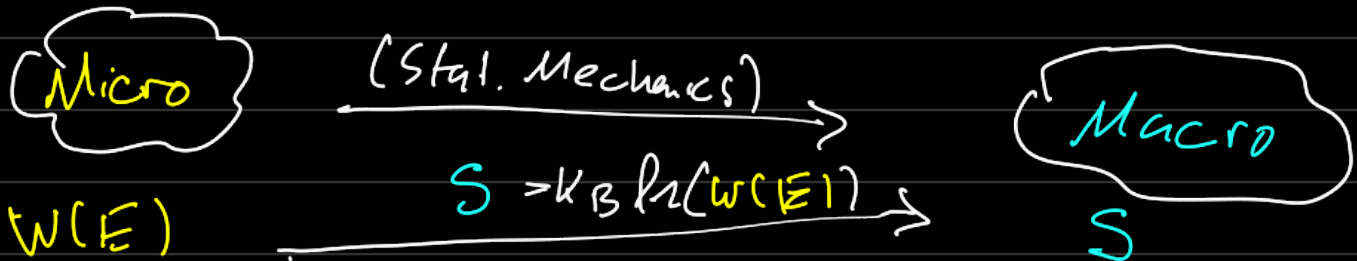
- Entropy: $S = k_B \ln(W(E))$

Entropy is a macroscopic term as we saw, S goes something like the total number of particles

$$S \sim N$$

This is something we can 'measure'. But we see it depends on a microscopic term $W(E)$ i.e. the (Quantum) micro states.

~> This is the first step in our overall goal



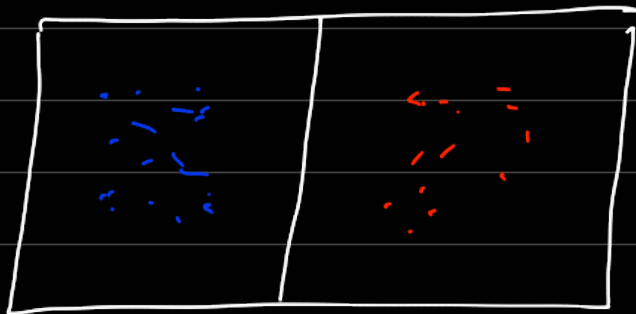
- In (the very boring) earlier sections of chps 1-4 you also looked at

G : Gibbs Energy
 F : Free energy
 U : (Average) Energy
 C : Heat capacity

all macro

So now that we know S and T we want to work towards (Quantum) Microscopic descriptions of the above terms. Before we do this lets understand how to work with what we just found.

[5.3 of your notes]



N_1, V

N_2, V

- System has N total particles split so that

$$N = N_1 + N_2$$

- Both systems have the same volume
- System similar to my prob. notes.

$$\binom{N}{N_1} = C(N, N_1) = \left(\begin{array}{l} \text{\# of ways of choosing} \\ N_1 \text{ objects from } N \text{ total} \end{array} \right)$$

- The number of micro states for N_1 :

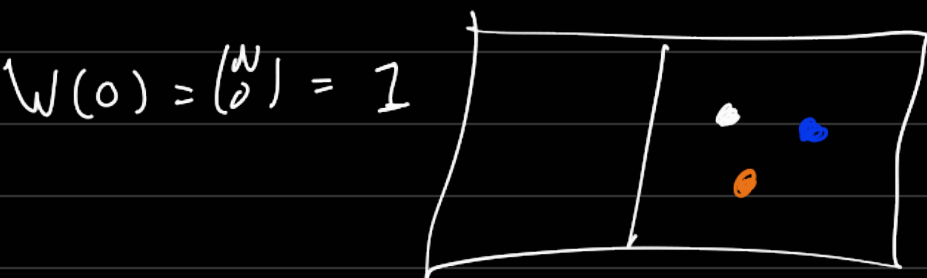
$$\binom{N}{N_1} = W(N_1) = \frac{N!}{N_1! (N - N_1)!}$$

Why is this? Well, it's a counting problem.

- Let $N = 3$, i.e. we have 8 different possibilities

$$N = 3$$

$$(a) \quad N_1 = 0, \text{ then } N_2 = 3$$



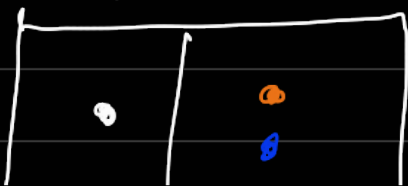
$$W(0) = \binom{3}{0} = 1$$

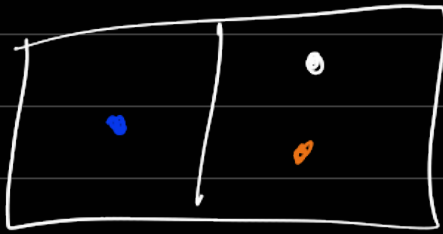
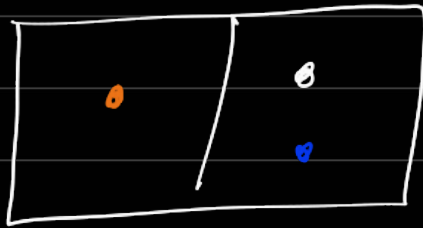
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There is only one way to do this

$$(b) \quad N = 3$$

$$N_1 = 1, N_2 = 2$$





3

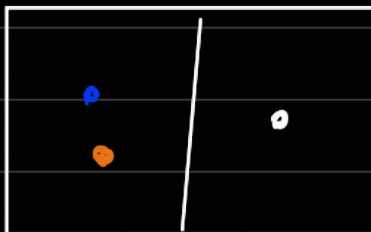
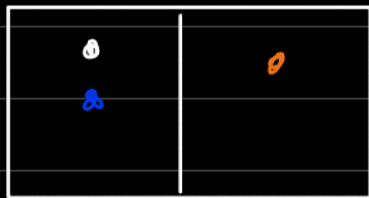
* Note there are three different possible ways to set this system up

$$W(1) = \binom{3}{1} = \binom{N}{N_1} = \frac{3!}{1!2!} = \frac{3 \cdot 2 \cdot 1}{2 \cdot 1} = \underline{3}$$

(c). For $N=3$

$$N_1 = 2, N_2 = 1$$

Again we have 3 possible outcomes

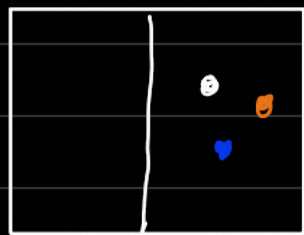


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$$W(2) = \binom{3}{2} = \frac{3!}{2!1!} = \underline{3}$$

(d) $N=3$

$$N_1 = 3, N_2 = 0$$



$$Q_1' = 1$$

(By definition)

$$W(3) = \binom{3}{3} = \frac{3!}{3!0!} = 1$$

• Thus, we see that the micro states for $W(N_1)$ is

$$W(N_1) = \binom{N}{N_1} = \frac{N!}{N_1!(N-N_1)!}$$

* Now, to understand the total microstates this is the sum of all possible N_1

$$W_{\text{tot}} = \sum_{n=1}^N W(n)$$

we know $W(n) = \binom{N}{n}$

$$W_{\text{tot}} = \sum_{n=1}^N \binom{N}{n}$$

• Okay, So how do we solve this? In my prob. notes I show how we have the Binomial Expansion

$$(p+q)^N = \sum_{n=1}^N \binom{N}{n} p^n q^{N-n}$$

Well, if we compare this to what we have we see that these two are equal for

$$p = q = 1:$$

$$(1+1)^N = 2^N = \sum_{n=1}^N \binom{N}{n} 1^n 1^{N-n} = \sum_{n=1}^N \binom{N}{n}$$

Thus $2^N = \sum_{n=1}^N \binom{N}{n}$

$$Z = \sum_{n=1}^{\infty} \binom{N}{n}$$

and so

$$W_{\text{tot}} = 2^N$$

as claimed in your notes. The prob. of having N_1 particles in the first volume is then

$$p(N_1) = \frac{W(N_1)}{W_{\text{tot}}} = \frac{\binom{N}{N_1}}{2^N}$$

Large Numbers:

- I mentioned how 10^{23} is quite large. But if $N = 10^{23}$ then

$$2^N \text{ and } N!$$

are unimaginably large. Note, for

ex: $N = 200 = 2 \times 10^2$

then

$$N! = 200! \sim 10^{49}$$

and

$$2^N = 2^{200} \sim 10^{60}$$

Now, even for only a "small" number like 200 2^N and $N!$ give a VERY large number. ... Now imagine for $N = 10^{23}$

Right, so, how do we deal with this? We look at so called Asymptotics (this is the mathematically rigorous way to look at this).

From this we find we can, to a very good approx take sums to integrals.

Why do we want to do this?

→ We have $P(N_1) = \frac{W(N_1)}{W_{\text{tot}}}$

But recall

$$S(N_1) = k_B \ln(W(N_1))$$

$$= k_B \ln(P(N_1) W_{\text{tot}})$$

$$S(N_1) = k_B \ln(P(N_1)) + k_B \ln(W_{\text{tot}})$$

$$\ln(P(N_1)) = \ln\left(\frac{N!}{2^N N_1! (N-N_1)!}\right)$$

$$= \ln(N!) - \ln(2^N N_1! (N-N_1)!)$$

- These are the types of terms which apply to the statement above. Since

$$N! = N \cdot (N-1) \cdot (N-2) \cdot \dots \cdot 2 \cdot 1$$

$$\ln(N!) = \ln(N) + \ln(N-1) + \dots + \ln(2) + \ln(1)$$

$$= \sum_{n=1}^N \ln(n)$$

or

$$\ln(N!) = \sum_{n=1}^N \ln(n)$$

again, the claim was, for very large #'s

$$\sum_{n=1}^N \ln(n) \longrightarrow \int_1^N \ln(n) dn$$

- This integral is straight forward:

$$\int_1^N \ln(n) dn = N(\ln(N) - 1)$$

so Stirling's approximation is given by

$$N! \approx \sqrt{2\pi N} \left(\frac{N}{e}\right)^N$$

$$\ln(N!) \approx N(\ln(N) - 1)$$

(for $N \gg 1$)

* Now, for sections 5.3.2 - 5.3.3 I don't believe these sections teach you much in terms of understanding. So, instead let's look at applying our new ideas.

[Similar to 5.5.1]

• We know the Entropy associated with the microstates $W(N_1)$ is

$$S(N_1) = k_B \ln(W(N_1))$$

$$S(N_1) = k_B \ln \left(\frac{N!}{N_1! (N-N_1)!} \right)$$

$$= k_B \left[\ln(N!) - \ln(N_1!) - \ln(N-N_1)! \right]$$

• Now, apply Sterling's approx. to each of these

$$\ln(N!) \approx N(\ln(N) - 1)$$

$$\ln(N_1!) \approx N_1(\ln(N_1) - 1)$$

$$\ln(N-N_1)! \approx (N-N_1)(\ln(N-N_1) - 1)$$

Plug this in:

$$S \approx k_B \left[N(\ln(N) - 1) - N_1(\ln(N_1) - 1) - (N-N_1)(\ln(N-N_1) - 1) \right]$$

$$\frac{S}{k_B} \approx \left[\underline{N} \ln(N) - \underline{N} \ln(N-N_1) + \underline{N_1} \ln(N-N_1) - \underline{N_1} \ln(N_1) - N + N_1 + (N-N_1) \right]$$

• We collected N and N₁ terms so that

$$\frac{S}{k_B} \approx \left[N \ln \left(\frac{N}{N-N_1} \right) + N_1 \ln \left(\frac{N-N_1}{N_1} \right) \right]$$

$$\star S(N_1) \approx k_B \left(N \ln \left(\frac{N}{N-N_1} \right) + N_1 \ln \left(\frac{N-N_1}{N_1} \right) \right) \star$$

Now, in our session our claim was that in equilibrium the most probable state was when $W(N_1)$ was maximal

This is equivalent to maximizing S since

$$\frac{\partial S}{\partial E} = k_B \frac{1}{W_B} \frac{\partial W_B}{\partial E} \stackrel{!}{=} 0$$

- Here, we see that we are working with N_1 , each which have some energy E . So we are looking at

$$\frac{\partial S}{\partial E} = 0$$

but by the chain rule

$$\frac{\partial S}{\partial E} = \frac{\partial N}{\partial E} \frac{\partial S}{\partial N} = 0$$

↳

Some number

so we can look at $\frac{\partial S}{\partial N_1} = 0$

Thermodynamics, as we saw, states that the most probable state is occupied at equilibrium

$$\frac{\partial S}{\partial N_1} = 0 = k_B \frac{\partial}{\partial N_1} \left(N \ln \left(\frac{N}{N-N_1} \right) + N_1 \ln \left(\frac{N-N_1}{N_1} \right) \right)$$

after a bit of algebra you'll find

$$(N_1)_* = N/2$$

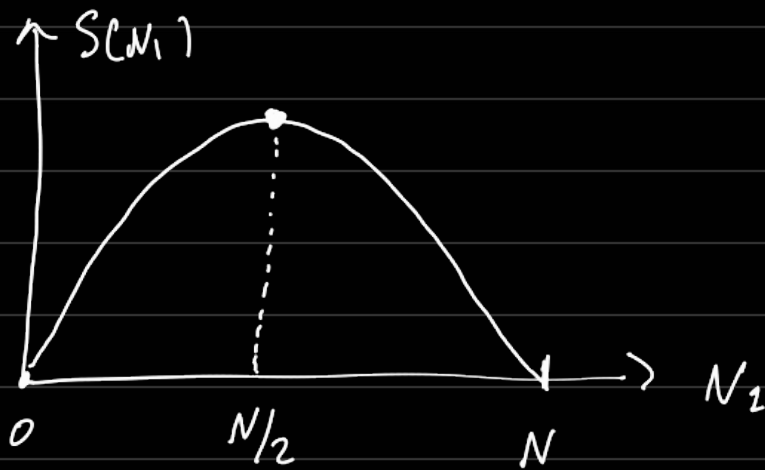
(Equilibrium / Most prob. State)

→ This is where the claim on page 5.3 [5.3.2] comes from where they say "Thermodynamics suggests equilibrium at $N_1 = N_2 = N/2$ "

- So, we've found that at Eq. the particles

will be evenly distributed as we might have guessed

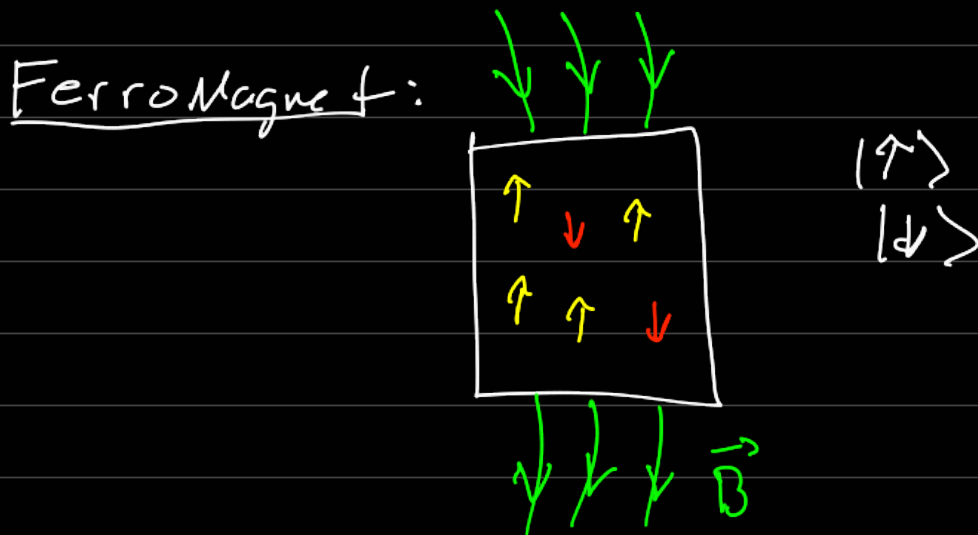
- If you sketch $S(N_1)$ you'll find



* We've done most of the work already to look at a two-level system
[5.5.1] - [5.5.2] so let's look at that now

- Energies: 0 or E

How can we interpret this? Well, instead of a lattice (as your notes state) let's think about the Quantum Spin of a particle. We are, after all, trying to relate the Quantum world to the one we experience.



Suppose we have a magnet whose elements have either spin $\uparrow$ or $\downarrow$. Then, if we apply an external magnetic field we know from QM / E&M that the spin which $\downarrow$ 'aligns' with B is unaffected while those which are anti-parallel will flip or align with the field.

In this way

$\downarrow$: has $E = 0$

i.e. no energy is required to align.

$\uparrow$: $E = E$:

i.e. the magnetic field will do work to flip this over. That work has an energy input of $-E$.

* Okay, we have a real model (ideal...but real)
The counting is exactly the same:

$\uparrow\uparrow\downarrow\uparrow\dots\dots$

$$N_1 = N_{\uparrow}$$

$$N_2 = N_{\downarrow}$$

$$N = N_{\uparrow} + N_{\downarrow}$$

$$\underline{N=3} : N_{\uparrow} = 3, N_{\downarrow} = 0$$

$\uparrow\uparrow\uparrow$

$$N_{\uparrow} = 2, N_{\downarrow} = 1$$

$\uparrow\uparrow\downarrow$

$\uparrow\downarrow\uparrow$

$\downarrow\uparrow\uparrow$

3

etc...

we saw this counting was simply

$$W(N_{\uparrow}) = \binom{N}{N_{\uparrow}} = \frac{N!}{N_{\uparrow}!(N-N_{\uparrow})!}$$

so as stated the set up is the same. The Energies:

$$E_{\text{tot}} = N_{\uparrow} E_{\uparrow} + N_{\downarrow} E_{\downarrow}$$

$$E_{\uparrow} = \epsilon, \quad E_{\downarrow} = 0$$

$$E_{\text{tot}} = N_{\uparrow} \epsilon$$

- Let's call $E_{\text{tot}} = E$, so that $N_{\uparrow} = \frac{E}{\epsilon}$
we can plug this into the entropy formula we found:

$$S(N_{\uparrow}) \approx k_B \left(N \ln \left(\frac{N}{N - N_{\uparrow}} \right) + N_{\uparrow} \ln \left(\frac{N - N_{\uparrow}}{N_{\uparrow}} \right) \right)$$

• Note again that our notation is $N_{\downarrow} = N - N_{\uparrow}$.

Replace $N_{\uparrow} = E/\epsilon$ (fractional Energy)

$$S(E) \approx k_B \left(N \ln \left(\frac{N}{N - (E/\epsilon)} \right) + \left(\frac{E}{\epsilon} \right) \ln \left(\frac{N - (E/\epsilon)}{(E/\epsilon)} \right) \right)$$

after some algebra you can get this in the form

$$S(E) \approx -k_B N \left(\left(1 - \frac{E}{N\epsilon} \right) \log \left(1 - \frac{E}{N\epsilon} \right) + \frac{E}{N\epsilon} \ln \left(\frac{E}{N\epsilon} \right) \right)$$

* Recall our maximum was at

$$N_{\uparrow} = N/2$$

which means this peaks at

$$N_{\uparrow} = E/\epsilon = N/2$$

or

$$E_* = \epsilon N/2$$

Plugging this in you'll find

$$S(E_*) = N k_B \ln(2) = S(W_{\text{tot}})$$

So, we made our equation appear a bit more

complicated. Why? Well, we know that

$$\left. \frac{\partial S}{\partial E} \right|_V = \frac{1}{T}$$

So to find the temperature we have to have some idea of how to relate the energy.

- Taking $\partial S / \partial E$ of the above:

$$\frac{\partial S}{\partial E} = \frac{k_B}{\epsilon} \ln \left(\frac{N\epsilon}{E} - 1 \right)$$

there's a lot of algebra but you'll find terms cancel out.

- Since $\partial S / \partial E = 1/T$

$$\frac{1}{T} = \frac{k_B}{\epsilon} \ln \left(\frac{N\epsilon}{E} - 1 \right)$$

* Solving for E :

$$\frac{\epsilon}{k_B T} = \ln \left(\frac{N\epsilon}{E} - 1 \right)$$

$$\Rightarrow e^{\epsilon/k_B T} = \frac{N\epsilon}{E} - 1$$

$$\frac{N\epsilon}{E} = 1 + e^{\epsilon/k_B T}$$

or

$$\boxed{\frac{E}{N\epsilon} = \frac{1}{1 + e^{\epsilon/k_B T}}}$$

- Note: Since $E = N_f \epsilon \Rightarrow N_f = \frac{E}{\epsilon}$
so

$$\boxed{N_f / N = \frac{E}{N\epsilon} = \frac{1}{1 + e^{\epsilon/k_B T}}}$$

- Again, let's nail two concepts in:

Equilibrium occurs at the max of S which we find by $\partial S / \partial E$. But

$$1 = \frac{\partial S}{\partial E} = \frac{1}{T}$$

$$\frac{\partial S}{\partial E} = \frac{1}{T} = 0$$

this means $T \rightarrow \infty$

• Indeed, For $T \rightarrow \infty$

$$\lim_{T \rightarrow \infty} \left(\frac{E}{k_B T} \right) = 0$$

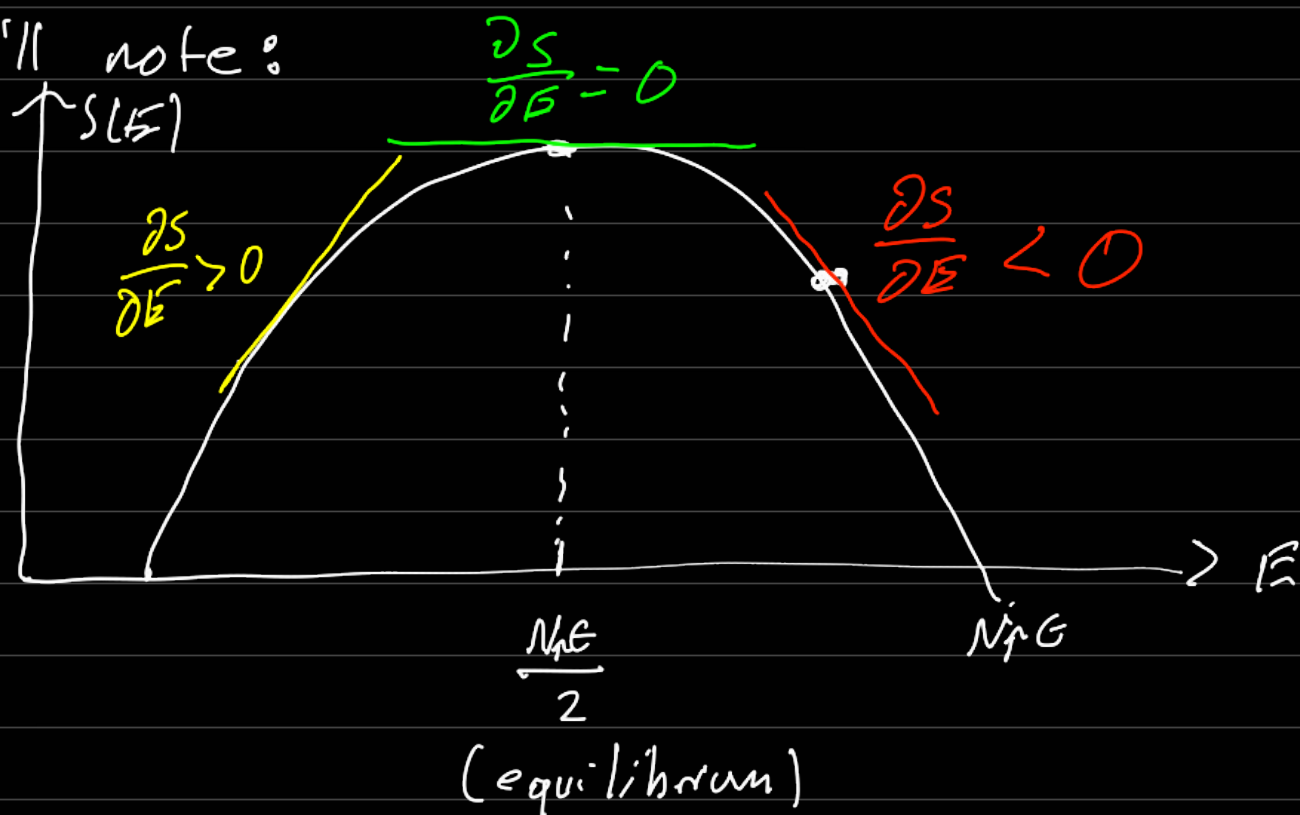
$$\text{so } \lim_{T \rightarrow \infty} \left(\frac{1}{1 + e^{E/k_B T}} \right) = \frac{1}{1 + e^0} = \frac{1}{2}$$

so

$$\frac{N_{\uparrow}}{N} = \frac{1}{2} \Rightarrow \boxed{N_{\uparrow} = \frac{N}{2}}$$

• This is exactly what we found above. So, the condition $T \rightarrow \infty$ tells us about the equilibrium state of the system!

* I'll note:



* Note there is a region where $\frac{\partial S}{\partial E} < 0$ i.e. it is negative. But then if

$$\frac{\partial S}{\partial E} = \frac{1}{T} \Rightarrow \frac{1}{T} < 0$$

or $T = -$ (Negative)

$1 < 0$ | temperatures: |

- Claim: $-T$ is HOTTER than infinite temp!
→ I want go into details but should look it up!

